

Augmented matrices

For the linear system

$$\begin{array}{cccccc} a_{11}x_1 + a_{12}x_2 + & \dots & + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + & \dots & + a_{2n}x_n & = & b_2 \\ \vdots & \ddots & \vdots & & \\ a_{m1}x_1 + a_{m2}x_2 + & \dots & + a_{mn}x_n & = & b_m \end{array}$$

the augmented matrix is

$$\left(\begin{array}{ccccc} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \ddots & \vdots & & \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

Ex: Solve

$$\rightarrow \left. \begin{array}{rcl} & 2y + 2z = -4 \\ 2x + 6y + 2z = 4 \\ 3x & & + z = 3 \end{array} \right\} \begin{array}{l} \text{syst.} \\ \text{of} \\ \text{lin.} \\ \text{equ.} \end{array}$$

Find all (x, y, z) that satisfy these equations.

Augmented matrix

$$\left(\begin{array}{ccc|c} 0 & 2 & 2 & -4 \\ 2 & 6 & 2 & 4 \\ 3 & 0 & 1 & 3 \end{array} \right)$$

3×4
matrix

Coeff. matrix

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 2 \\ 3 & 0 & 1 \end{pmatrix} \leftarrow 3 \times 3 \text{ matrix.}$$

$$\underline{2x} + 6y + 2z = 4$$

$$2y + 2z = -4$$

$$3x + z = 3$$

Interchanged
rows 1 and
2.

Divide row 1 by 2.

$$x + 3y + z = 2$$

$$2y + 2z = -4$$

$$+ z = 3$$

3x

-3 × row 1 + row 3 (and place in row 3)

$$\underline{x} + 3y + z = 2$$

$$\underline{2y} + 2z = -4$$

$$-9y - 2z = -3$$

$$\begin{array}{rcl}
 x + 3y + z & = & 2 \\
 y + z & = & -2 \\
 \underline{-9y + 2z} & = & -3 \quad ||
 \end{array}$$

9 $\times$ row 2 + row 3 $\leadsto$ put in row 3

$$x + 3y + z = 2$$

$$\begin{array}{rcl}
 y + z & = & -2 \\
 \hline
 7z & = & -21 \leadsto z = -3
 \end{array}$$

So $z = -3$, $y = 1$ and $x = 2$.

(Check this is a solution to the original system.)

Ex: Using augmented matrices solve

$$\left(\begin{array}{ccc|c} -1 & 2 & 0 & 3 \\ \frac{1}{2} & 1 & -2 & 4 \\ 1 & 3 & -2 & 8 \end{array} \right)$$

$$-x + 2y = 3$$

$$2x + y - 2z = 4$$

$$x + 3y - 2z = 8$$

$$\hookrightarrow \left(\begin{array}{ccc|c} \textcircled{1} & -2 & 0 & -3 \\ \textcircled{2} & 1 & -2 & 4 \\ 1 & 3 & -2 & 8 \end{array} \right)$$

← mult. row 1 $\times (-1)$

$$\hookrightarrow \left(\begin{array}{ccc|c} 1 & -2 & 0 & -3 \\ 0 & 5 & -2 & 10 \\ 0 & 5 & -2 & 11 \end{array} \right)$$

$$-2 \times \text{row 1} + \text{row 2}$$

$$-\text{row 1} + \text{row 3}$$

No solution!

If we took $-1 \times \text{row 2} + \text{row 3}$ we get

$$\begin{pmatrix} 1 & -2 & 0 & -3 \\ 0 & 5 & -2 & 10 \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix} \leftarrow 0 = 1 \neq.$$

Elementary row operations

To solve a linear system one performs a series of algebraic operations:

- Interchange equations
- Multiply an equation by a non-zero constant
- Add a constant multiple of one equation to another

The main fact is that doing these operations doesn't change the set of solutions of the linear system.

The corresponding *elementary row operations* on an augmented matrix are:

- Interchange two rows
- Multiply a row by a non-zero constant
- Add a constant multiple of one row to another

Echelon forms

If a row in a matrix isn't entirely zeroes then the left most non-zero entry is called the *leading* entry.

A matrix is in **row echelon form** if

- 1 All zero rows are grouped contiguously at the bottom of the matrix.
- 2 In any two consecutive rows which are both non-zero, the leading entry of the upper row is to the left of the leading entry of the lower row.
- 3 If additionally, all leading entries are 1 and any column which contains a leading 1 has zeroes elsewhere, then we say the matrix is in **reduced row echelon form**.

Gaussian and Gauss-Jordan elimination

- 1 Given the augmented matrix for a system of linear equations, find the left most column which is not all zero.
- 2 Interchange rows so that the non-zero entry from the previous step is the top row.
- 3 If the non-zero entry you have found is a then divide the top row by a so that its left most entry is 1.
- 4 Work down row by row adding multiples of the first row to each row to guarantee that all entries below this 1 are zero.
- 5 Now ignore the top row and repeat the process with the rows you have remaining until there are no more rows left; this gets the matrix into row echelon form.
- 6 For reduced row echelon form, start with the right most leading 1 and working left, add suitable multiples of that row to the rows above to guarantee that all entries above leading 1's are 0.

Solving a system of equations by row-reducing the augmented matrix

- Start with the augmented matrix of a linear system and row-reduce it.
- If the right most column contains a leading 1 then the system has no solutions.
- Otherwise, the variables that correspond to columns with leading 1's - pivot columns - can be written in terms of the other variables, the free variables.
- If there is a solution and no free variables then the solution is unique.
- If there are solutions and free variables then there are infinitely many solutions.